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Clinical efficacy and safety of *Lawsonia inermis* (Henna): A systematic review of human clinical studies

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ABSTRACT

Background: Lawsonia inermis (henna) is used for diverse clinical conditions, but its benefits and harms in humans remain uncertain.
Objective: To assess the clinical efficacy and safety of henna-containing interventions.
Methods: PubMed, Embase, and Scopus were searched on 1 October 2025 and updated on 19 September 2026, when CENTRAL was added. A targeted Google Scholar author search identified one additional eligible report. Randomized, nonrandomized comparative, and single-arm studies were synthesized narratively because interventions and outcomes were heterogeneous. Risk of bias was assessed with RoB 2, ROBINS-I, or the NIH tool; certainty was assessed with GRADE. The review was registered in PROSPERO (CRD420251155578).
Results: Twenty-five reports represented 24 unique studies. Thirteen studies evaluated henna as the sole stated active component, nine evaluated multicomponent formulations, and two evaluated henna under shared active treatment. Only 12 studies described ongoing adverse-event or complication surveillance, and safety denominators were often unclear. Reported events included an allergic reaction leading to exclusion, local pruritus, and withdrawals associated with skin reactions. One trial reported a death among noncompleters without stating its cause or relationship to treatment. Selected comparative outcomes favoured henna or a henna-containing regimen, but findings varied by condition and comparator; effects of multicomponent formulations cannot be attributed to henna alone. Certainty was low for seven outcome groups and very low for 12.
Conclusion: The evidence does not establish general safety or reliable efficacy across indications. Henna-containing interventions should not replace established care. Better-reported comparative trials with systematic safety monitoring are needed.

1. Introduction

Lawsonia inermis L., commonly known as henna, is a herbal agent that has long been used in the Middle East, South Asia, and North Africa. Traditionally, henna has been used in cultural and religious celebrations as a natural dye for the hair, skin, and nails. Beyond its cosmetic role, it has also held an important place in traditional medicine, where it has been applied to treat burns, wounds, infections, and pain.[1] This deep cultural and historical usage underpins modern interest in henna's medicinal potential.

Phytochemical studies have revealed several bioactive constituents in henna that may contribute to its therapeutic effects. The most prominent is lawsone (2-hydroxy-1,4-naphthoquinone), a red-orange pigment with notable anti-inflammatory and analgesic properties.[1,2] Henna also contains flavonoids, tannins, and other phenolic compounds with antioxidant and antimicrobial activity, supporting its traditional use in managing infection and inflammation.[2] While early clinical studies suggest potential benefits in conditions such as contact dermatitis, wound healing, and pain management, robust evidence from high-quality trials remains limited.

Many of the conditions for which henna has traditionally been used, such as chronic wounds, inflammatory skin diseases, infections, and pain, represent a substantial and growing global health burden. Chronic wounds alone affect an estimated 1–2 % of the population in developed countries[3–6] and are responsible for tens of billions of dollars in annual healthcare costs.[7] Conventional treatments for these conditions, such as corticosteroids, nonsteroidal anti-inflammatory drugs (NSAIDs), and antibiotics, are not only costly but also linked to a broad spectrum of

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adverse effects affecting the musculoskeletal, endocrine, and gastrointestinal systems.[8] Notably, long-term corticosteroid use has been associated with cutaneous side effects, including skin atrophy.[9] In addition, access to modern medications remains limited in many low-income and rural regions, with nearly 2 billion people worldwide lacking access to essential medicines.[10] These challenges highlight the growing interest in alternative therapies that are affordable, culturally accepted, and plant-based, such as henna.

Pharmacological investigations have begun to clarify the pathways through which *Lawsonia inermis* exerts its effects. Lawsone, the plant's primary pigment, along with flavonoids and tannins, has been shown to suppress pro-inflammatory mediators such as TNF-α, IL-1β, and NF-κB in preclinical models, supporting its potential for anti-inflammatory and wound-healing applications.[11,12] Clinical studies, though still limited, have reported encouraging outcomes. A randomized controlled trial demonstrated that topical henna significantly improved symptoms of contact dermatitis in prosthetic users compared to placebo,[13] while another trial found it effective in preventing pressure ulcers among critically ill patients.[14] Despite these findings, not all studies report consistent results, which may be due to differences in henna preparation, concentration, dosing regimens, or study design.

A substantial portion of the available evidence still derives from in vitro and animal studies, with relatively few well-designed human clinical trials to date. Moreover, variability in extraction methods and formulation procedures further contributes to the inconsistency across findings.[15] While one recent review explored the anti-inflammatory properties of henna, it focused primarily on experimental models and did not comprehensively assess clinical outcomes or broader therapeutic applications in humans.[11]

In light of these gaps, this article presents a systematic review of the clinical efficacy and safety of henna-containing interventions. By synthesizing clinical evidence, the review aims to identify areas where henna demonstrates therapeutic potential, clarify where findings remain inconclusive, and assess its relevance in current medical practice. The results are intended to support clinicians, researchers, and policymakers in evaluating henna as a complementary treatment option.

2. Traditional plant-based treatments for target conditions

A wide range of phytotherapies have been used across ethnomedical systems for the management of conditions, including those evaluated in this review. These include wound healing conditions (episiotomy wounds, pressure ulcers, and pilonidal sinus disease), inflammatory skin disorders (dermatitis and hand-foot syndrome), and mucosal conditions (vaginal candidiasis and cervicitis). These traditional therapies frequently overlap in phytochemical profiles and share pharmacological mechanisms.

In the context of wound healing and burns, plants such as *Aloe vera* (L.) Burm.f., *Curcuma longa* L. (turmeric), and *Azadirachta indica* A.Juss. (neem) have been extensively used in traditional medicine systems. *Aloe vera* contains polysaccharides and glycoproteins that enhance fibroblast proliferation, collagen deposition, and re-epithelialization, accelerating wound closure.[16] Turmeric is rich in curcuminoids, which exhibit potent anti-inflammatory and antioxidant activity through the modulation of NF-κB and cytokine signalling pathways.[17] Neem contains limonoids and flavonoids with antimicrobial and immunomodulatory properties that support infection control and tissue repair.[18] These mechanisms align with those proposed for *Lawsonia inermis*, which has demonstrated antimicrobial activity, modulation of inflammatory mediators, and promotion of wound healing in both preclinical and clinical settings.[1,12]

For inflammatory skin conditions, including dermatitis and treatment-related skin toxicities such as hand-foot syndrome, traditional remedies include *Calendula officinalis* L. and *Matricaria chamomilla* L. These agents contain flavonoids, terpenoids, and phenolic compounds that have anti-inflammatory effects through the inhibition of cyclooxygenase and the reduction of pro-inflammatory cytokines.[19,20] Comparable mechanisms have been described for *Lawsonia inermis*, where its phenol and lawsone content contribute to anti-inflammatory and antioxidant activity, supporting its use in dermatitis and inflammatory skin conditions.[2]

In the management of infectious and mucosal conditions, including candidiasis and cervicitis, traditional therapies include *Melaleuca alternifolia* (Maiden & Betche) Cheel (tea tree) and *Allium sativum* L. (garlic), both of which display antimicrobial activity. Tea tree oil contains terpinen-4-ol and other terpenes that disrupt microbial cell membranes, while allicin derived from garlic exerts antimicrobial effects by inhibiting key enzyme systems in pathogens.[21,22] Similarly, lawsone, the principal compound of *Lawsonia inermis*, has demonstrated antibacterial and antifungal activity, providing a basis for its use in infectious conditions.[23]

Additionally, plant-based therapies such as *Zingiber officinale* Roscoe (ginger) and *Curcuma longa* (turmeric) have traditionally been used for pain and inflammation, acting through the inhibition of inflammatory mediators including TNF-α, IL-1β, and NF-kB.[17,24] These mechanisms are consistent with experimental findings demonstrating that lawsone and other constituents of *Lawsonia inermis* modulate similar inflammatory pathways.[12]

3. Methods

3.1. Protocol registration and reporting guidelines

This systematic review protocol was registered in the PROSPERO database (registration number: CRD420251155578). The review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to ensure methodological transparency and rigor.[25]

Following the original search, CENTRAL was added to the search update conducted on 19 September 2026 to improve coverage of controlled trials. Although the protocol proposed meta-analysis where studies reported sufficiently comparable outcomes, pooling was judged inappropriate during final synthesis because interventions, comparators, outcome definitions, and follow-up differed substantially. Results were therefore synthesized narratively. For synthesis, interventions were also grouped as henna alone, multicomponent formulations, or henna under shared active treatment to distinguish which effects could be attributed to henna.

3.2. Literature search

Given the focus of this systematic review on the clinical efficacy and safety of *Lawsonia inermis* (henna) across medical contexts, a comprehensive literature search was undertaken. PubMed, Embase, and Scopus were searched because of their extensive coverage of biomedical, pharmacological, and clinical research.

The review team developed and refined the database-specific search strategies. The complete database-specific search strategies are provided in Supplementary Table S1.

To enhance sensitivity and minimize the risk of missing relevant evidence, the search strategy was deliberately broad rather than narrowly restrictive. No date restrictions were imposed.

PubMed, Embase, and Scopus were searched on 1 October 2025. The search was updated in these three databases and extended to the Cochrane Central Register of Controlled Trials (CENTRAL) on 19 September 2026. Complete database-specific strategies, search dates, and record counts for both searches are provided in Supplementary Table S1. Retrieved records were imported into Covidence, where duplicate citations were identified and removed prior to screening.

Following identification of a conference abstract by Lesan et al. during screening,[26] the Google Scholar author profile of M. Lesan was consulted on 3 October 2026 for subsequent publications with henna in the title. This targeted search identified the corresponding full-text

publication by Kamian et al.[27]), which had not been retrieved through the database searches.

3.3. Inclusion and exclusion criteria

The eligibility criteria for this systematic review were developed using the PICO (Population, Intervention, Comparator, and Outcomes) framework, with study design also considered.

Studies were included if they (1) involved human participants of any age, (2) evaluated the clinical use of *Lawsonia inermis* (henna) for therapeutic or prophylactic purposes in any clinical or medical context, (3) investigated henna administered in any formulation or route, including topical, oral, extract-based, paste, oil, or compound preparations, whether used alone or in combination with other substances, (4) included a comparator group where applicable (e.g., placebo, no treatment, standard care, or an active comparator) or employed a single-arm clinical design with clearly defined clinical outcomes, (5) reported clinically relevant efficacy outcomes (e.g., healing rate, lesion size, symptom relief, inflammation reduction, or quality of life) and/or safety outcomes including adverse events, and (6) employed a human clinical study design, including randomized controlled trials, non-randomized trials, single-arm trials, or other controlled clinical studies. Only full-text, peer-reviewed articles published in English were considered eligible.

Studies were excluded if they (1) were conducted in non-human, animal, or *in vitro* models, (2) involved healthy participants using henna solely for non-clinical or cosmetic purposes, (3) evaluated henna exclusively as a cosmetic dye or in non-therapeutic contexts, (4) investigated multi-component formulations in which *Lawsonia inermis* was not an identifiable ingredient or in which the clinical contribution of a henna-containing formulation could not be evaluated, (5) employed non-clinical or inappropriate study designs such as case reports, case series, narrative reviews, systematic reviews, conference abstracts, dissertations, or other grey literature, or (6) were not published in English or available as full-text articles.

Eligible interventions were classified before synthesis as (A) henna as the sole stated active component, (B) a multicomponent henna-containing formulation, or (C) henna evaluated while both groups received a shared active treatment. Category B provides formulation-level evidence. Category C permits a comparison under the shared treatment, but does not establish the effect of henna used alone.

3.4. Data Screening

Two reviewers independently screened titles and abstracts in Covidence against the eligibility criteria; disagreements were resolved through discussion and, when needed, consultation with the review team. Full-texts were independently assessed by two reviewers. The original search yielded 21 eligible full-text reports, all of which were included. After the September 2026 update, 23 additional full-text reports were assessed, 20 were excluded, and three were included. One additional full-text report was located through a targeted Google Scholar follow-up using author and title details from a conference abstract identified during screening and was assessed against the same eligibility criteria. In total, 25 reports represented 24 unique studies; the Poursadra and Rafiei reports described overlapping cohorts and were treated as one linked study family.[28,29] Fig. 1 shows each search stage and the updated full-text exclusion reasons.

3.5. Quality appraisal

Risk of bias was assessed using the Cochrane Risk of Bias 2 (RoB 2)

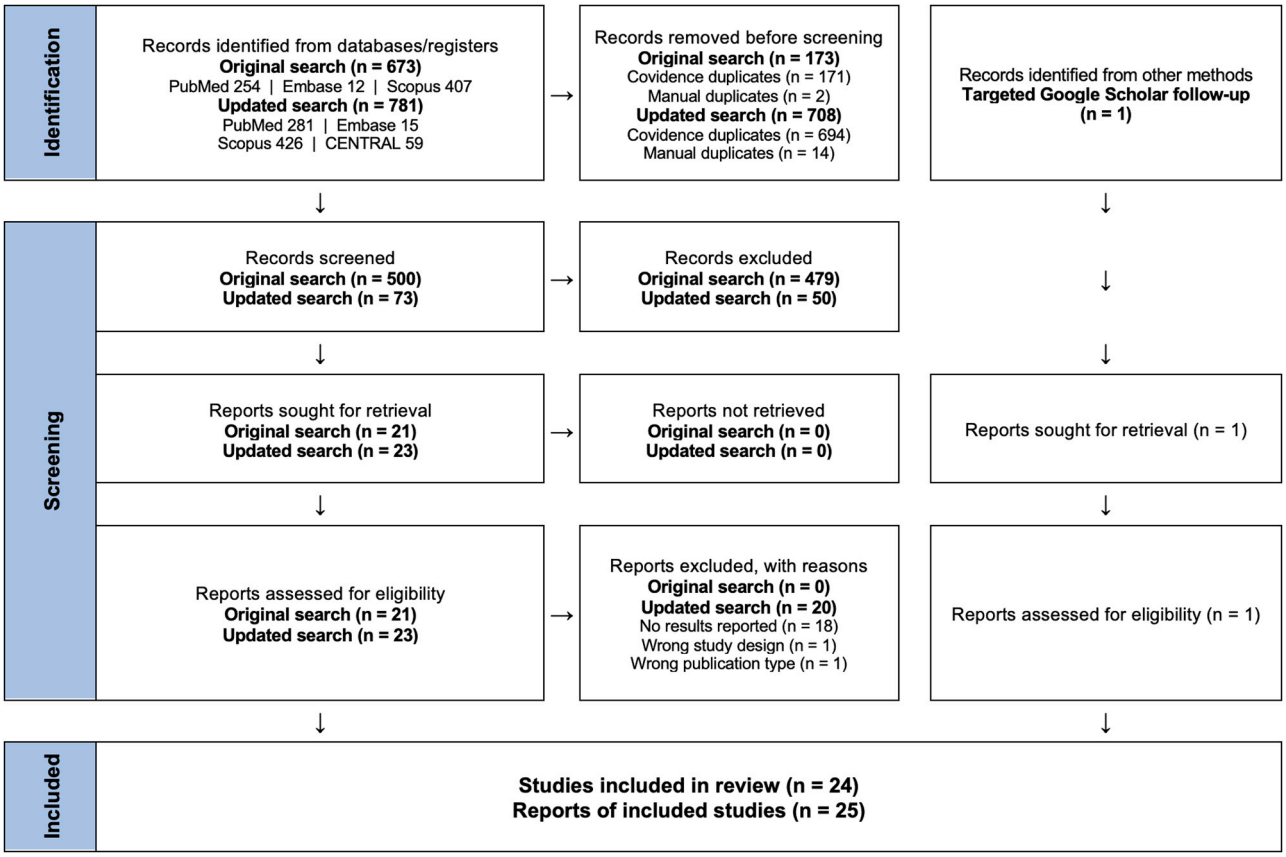


Fig. 1. Study selection. PRISMA 2020 flow diagram showing identification, screening, full-text assessment, and inclusion across the original and updated searches. The review includes 25 reports of 24 unique studies.

tool for randomized controlled trials and the Risk of Bias in Non-randomized Studies of Interventions (ROBINS-I) tool for non-randomized studies.[30,31] For single-arm clinical trials, methodological quality was evaluated using the National Institutes of Health (NIH) Quality Assessment Tool.[32]

Two reviewers independently applied the relevant appraisal tool, and disagreements were resolved by consensus. RoB 2 judgments were made for 22 randomized-trial assessments, ROBINS-I for two non-randomized comparisons, and the NIH tool for one single-arm study. The linked Poursadra/Rafiei study family contributed to both a randomized and a nonrandomized comparison; therefore, tool-specific assessment counts should not be summed as independent studies.

Risk-of-bias judgments were subsequently incorporated into the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework to inform the overall certainty of the evidence.[33] Domain-level and overall judgments across randomized, non-randomized, and single-arm studies are summarized in Supplementary Figures S2–S6.

3.6. Data extraction and synthesis

Study data were extracted independently by two reviewers using a predefined extraction template. Information collected included study design and setting, participant characteristics, details of the henna intervention and any comparator, clinical condition assessed, outcome measures, numerical results, reported safety data, and risk-of-bias evaluations. Detailed descriptions of intervention and comparator formulations, including concentrations, routes of administration, and accompanying standard or active treatments, are provided in Supplementary Table S7. Numerical study outcomes are provided in Supplementary Table S8. Additionally, botanical names of all plant species mentioned in this review were verified using the Medicinal Plant Names Services (MPNS) database,[34] and are reported with their corresponding authorities at first mention (accessed 2 April 2026).

Statistical pooling was not undertaken because outcomes were reported by a small number of studies (typically one or two per outcome), and outcome definitions and assessment tools varied substantially across studies. As a result, evidence was synthesized qualitatively using a structured narrative approach, informed by Cochrane guidance for narrative synthesis (Chapter 12) and reported in accordance with the Synthesis Without Meta-analysis (SWiM) guidelines.[35,36]

Studies were grouped by clinical condition, outcome, and intervention category. Direction-of-effect judgments for controlled studies were based on reported between-group comparisons for clinically relevant outcomes, with effect estimates and confidence intervals used when available. A within-group change or isolated P value did not establish comparative benefit. When no between-group test or effect estimate was available, the comparative direction was recorded as not estimable. Single-arm findings were described separately and were not used to infer comparative efficacy. No pooled effect was calculated because interventions, comparators, outcome definitions, and follow-up differed substantially.

The strength of evidence for each outcome group was subsequently appraised using the GRADE framework,[33] with judgments based on the five recognized downgrading domains: risk of bias, inconsistency, indirectness, imprecision, and publication bias. Certainty ratings were used to contextualize the findings and support interpretation of the results. Final GRADE judgments for each outcome are summarized in Table 3 and Supplementary Table S9.

4. Results

4.1. Study selection

The original search identified 673 records (PubMed, n = 254; Embase, n = 12; Scopus, n = 407). After 173 duplicate records were removed, 500 records were screened; 479 were excluded and 21 full-text reports were included. The updated search identified 781 records (PubMed, n = 281; Embase, n = 15; Scopus, n = 426; CENTRAL, n = 59). After removal of 708 duplicate records against the existing and updated set, 73 additional records were screened; 50 were excluded, 23 full-text reports were assessed, and 20 were excluded (18 reported no results, one had an ineligible study design, and one had an ineligible publication type). Three additional reports were included. One further eligible full-text report was identified through a targeted Google Scholar author search prompted by a conference abstract and was included outside the database-search pathway. The final review contains 25 reports representing 24 unique studies (Fig. 1). The report by Yucel and Guzin on topical henna for capecitabine-induced hand–foot syndrome was excluded because it described an uncontrolled case series of ten patients and therefore did not meet the predefined eligible study-design criteria.[37]

4.2. Study characteristics

Table 1 summarizes 24 unique studies reported in 25 publications. The studies were concentrated in Iran (n = 18), followed by Turkey (n = 2), India (n = 2), Iraq (n = 1), and Cyprus (n = 1). The geographical distribution of studies is summarized in Fig. 2.

Thirteen studies (14 reports) evaluated henna as the sole stated active component, nine assessed multicomponent henna-containing formulations, and two compared henna under a shared active cointervention (Supplementary Table S7).

Most designs included a randomized comparison; Kayikci used a nonrandomized within-person comparison,[43] Niazi (2022) was single-arm,[48] and the linked Poursadra/Rafiei reports contributed randomized and nonrandomized comparisons.[28,29] Study-specific enrollment and analysis denominators are presented in Table 1; an aggregate participant total is not reported because the linked reports overlap and the Bhise and Khatami studies contain inconsistent participant counts.[55,57]

4.3. Risk of bias

Across 22 RoB 2 randomized-trial assessments, seven were rated as low risk of bias, eight had some concerns, and seven were rated as high risk of bias. The two ROBINS-I comparisons were judged to have serious risk of bias (Kayikci) and moderate risk of bias (Poursadra), respectively.[28,43] The Niazi (2022) single-arm study was rated as fair quality using the NIH tool.[48] These tool-specific assessments are reported in Table 2 and Supplementary Figures S2–S6; the linked Poursadra/Rafiei family is not counted twice as a unique study.[28,29]

4.4. Synthesis of results

Table 2 presents the study-level comparative synthesis, safety reporting, and risk-of-bias assessments. Detailed numerical results are provided in Supplementary Table S8. Table 3 summarizes outcome-level direction of effect and GRADE certainty.

4.4.1. Hand-foot syndrome (HFS)

Four studies reported hand-foot syndrome or palmar–plantar erythrodysesthesia outcomes. Neither prevention comparison established a clear reduction in incidence; Alpha ointment suggested delayed severity progression but is multicomponent.[41,43] Neither severity trial showed a clear between-group difference in clinical grade.[46,51] Stavrinou nevertheless reported better pain, function, fatigue, and quality-of-life measures with henna.[51] These symptom outcomes should be distinguished from the nonsignificant clinical-grade result (Table 2; Supplementary Table S8).

4.4.2. Dermatitis

Six studies assessed dermatitis-related outcomes. The

Table 1
Characteristics of the included studies.

Author (year)	Country	Study Design	Population Description	Sample Size	Duration	Intervention	Comparator	Location
Ansari et al.,[38]	Iran	RCT	Adults with locally advanced breast cancer and radiotherapy-induced dermatitis	60 (30 vs 30)	3 weeks	Alpha® ointment containing lawsone/natural henna and unsaturated fatty acids	Topical corticosteroid (hydrocortisone)	Chest wall (radiodermatitis)
Arslan et al.,[39]	Turkey	RCT	Women receiving oxaliplatin-based chemotherapy with chemotherapy-induced peripheral neuropathy	60 (30 vs 30)	45 days	Topical Lawsonia inermis (henna) powder	Routine care	Palms, fingers, soles, and toes
Dorbati et al.,[40]	Iran	RCT	Primiparous women with episiotomy	128 (64 vs 64)	1 week	Alpha® ointment with standard care	Betadine® sitz bath	Episiotomy site
Elyasi et al.,[41]	Iran	RCT	Adults with gastrointestinal or breast cancer receiving capecitabine for the first time	110 randomized; 90 analyzed (46 vs 44)	12 weeks	Alpha® ointment (henna-based topical preparation)	Placebo ointment	Palms and soles
(Hekmatpou et al., 2018)	Iran	RCT	Adult ICU patients at risk of pressure ulcers	80 randomized; 72 analyzed (37 vs 35)	1 week	Topical Lawsonia inermis preparation with standard care	Standard pressure-ulcer care	Sacrum
Kakoei et al.,[42]	Iran	RCT	Patients with symptomatic oral lichen planus	36 (18 vs 18)	2 weeks	Henna-based mouthwash (adjunct to standard therapy)	Chlorhexidine mouthwash with topical corticosteroid	Oral cavity (rinse)
Kayikci et al.,[43]	Turkey	Non-RCT	Breast or colorectal cancer patients receiving capecitabine	70 enrolled; 52 completed/ analyzed	At least three capecitabine cycles, with weekly application and follow-up	Topical Lawsonia inermis aqueous preparation	Contralateral untreated hand and foot with standard clinical care	One hand and one foot, with contralateral within-person control
Keshavarz et al.,[44]	Iran	RCT	Infants and young children with diaper dermatitis	82 (41 vs 41)	5 days	Topical henna oil-based cream	Topical corticosteroid (hydrocortisone)	Diaper dermatitis area
Khosravi et al.,[45]	Iran	RCT	Patients with postoperative pilonidal sinus disease	63 recruited; 60 analyzed (20 vs 20 vs 20)	4 weeks	Topical Lawsonia inermis extract ointment	Active comparator (phenytoin) and placebo control	Pilonidal sinus (topical)
Mohajerani et al.,[46]	Iran	RCT	Adults with gastrointestinal or breast cancer and fluoropyrimidine-induced hand–foot syndrome	18 entered treatment; 9 participants analyzed	2 weeks	Topical Lawsonia inermis extract hydrogel	Placebo gel applied to the contralateral hand/foot	Opposite hands/ feet within each participant
Nabimeybodi et al.,[47]	Iran	RCT	Premenopausal women with cervicitis	92 randomized; 82 analyzed (41 vs 41)	2 weeks	Henna-based vaginal preparation with adjunct antibiotics	Placebo vaginal preparation with standard antibiotics	Vaginal (suppository)
Niazi et al.,[13]	Iran	RCT	Lower-limb amputees with prosthetic-related contact dermatitis	95 randomized; 74 analyzed (37 vs 37)	2 weeks	Topical Lawsonia inermis extract in emollient base	Eucerin vehicle placebo ointment	Contact dermatitis site
Niazi et al.,[48]	Iran	Non-RCT	Patients with recessive dystrophic epidermolysis bullosa	9 enrolled; 7 analyzed	4 weeks	Topical Lawsonia inermis ointment	No comparator (single-arm study)	Erosions and pruritic sites
Pasalar et al.,[49]	Iran	RCT	Women with breast cancer and radiotherapy-induced dermatitis	106 randomized (53 vs 53); 93 analyzed (46 vs 47)	4 weeks	Polyherbal topical formulation containing Lawsonia inermis	Topical corticosteroid (mometasone)	Radiodermatitis lesion
Poursadra et al.,[28,29]	Iran	Overlapping reports; randomized two-arm report with three-arm comparative extension	Adult ICU patients with grade-one pressure ulcers	108 (36 vs 36 vs 36)	1 week	Topical aqueous henna preparation	Standard skin care and topical olive oil	Pressure ulcers (scapula, sacrum, heel)
Salih et al.,[50]	Iraq	RCT	Adults with primary sacrococcygeal	400 (200 vs 200)	Mean follow-up 2 years (range 6	Injection of a multicomponent mixture containing	Surgical management (standard care)	Pilonidal sinus (injection)

(*continued on next page*)

Table 1 (*continued*)

Author (year)	Country	Study Design	Population Description	Sample Size	Duration	Intervention	Comparator	Location
			pilonidal sinus disease		months–5 years)	petroleum jelly, henna powder, and tetracycline		
Stavrinou et al.,[51]	Cyprus	RCT	Adults receiving chemotherapy with palmar–plantar erythrodysesthesia	56 (26 vs 30)	4 weeks	Topical Lawsonia inermis application	Inert botanical powder (placebo)	PPE-affected areas (henna)
Takkella et al.,[52]	India	RCT	Adults with chronic periodontitis	60 (20 vs 20 vs 20)	2 weeks	Henna leaf extract mouthwash	Distilled water and Punica granatum mouthwash	Oral cavity (mouthwash)
Yaralizadeh et al.,[53]	Iran	RCT	Women with Candida albicans vulvovaginal candidiasis	80 (40 vs 40)	4 weeks	Henna-based vaginal cream	Topical antifungal (clotrimazole)	Vaginal (topical)
Zibanejad et al.,[54]	Iran	RCT	Primiparous women undergoing episiotomy	160 (40 vs 40 vs 40 vs 40)	2 weeks	Topical Lawsonia inermis ointment	Active comparator (oak ointment), emollient, or no intervention	Episiotomy site (topical)
Bhise et al.,[55]	India	RCT	Adults with uncomplicated psoriasis vulgaris	65 enrolled; analysis N NR[a]	3 months	Polyherbal topical formulation containing Lawsonia inermis	Topical corticosteroid (clobetasol)	Psoriatic skin lesions (topical)
Lavari et al.,[56]	Iran	RCT	Adults with chronic sciatica	81 (27 vs 27 vs 27)	4 weeks	Topical Lawsonia inermis extract in sesame oil	Sesame oil and liquid paraffin placebo	Lower back and painful leg
Khatami et al.,[57]	Iran	RCT	Adults with mild-to-moderate contact dermatitis	64 randomized; 62 analyzed (30 vs 32)[b]	4 weeks	Polyherbal cream containing Curcuma longa, Lawsonia inermis, and Aristolochia rotunda	Placebo base cream	Contact dermatitis sites (topical)
Kamian et al.,[27]	Iran	RCT	Women with breast cancer receiving radiotherapy after breast-conserving surgery	50 randomized; 43 analyzed (20 vs 23)	6 weeks	Topical henna–violet cream	Matched placebo cream	Irradiated breast skin

Notes: Data are presented as total N (group allocation) unless otherwise indicated. [a]Bhise (2022): Outcome-specific sample sizes were not reported, and participant and attrition counts were internally inconsistent. [b]Khatami (2026): The Results section and baseline table report 62 analyzed participants (30 CLA and 32 control), whereas the CONSORT diagram reports 60 participants (30 per group). Kayikci (2020) used a within-person controlled design, and Niazi (2022) used a single-arm design. Poursadra and Rafiei reported overlapping henna and control cohorts and were treated as linked reports of one study.
Abbreviations: CLA, Curcuma longa, Lawsonia inermis, and Aristolochia rotunda; CONSORT, Consolidated Standards of Reporting Trials; ICU, intensive care unit; N, number of participants; NR, not reported; PPE, palmar-plantar erythrodysesthesia; RCT, randomized controlled trial.

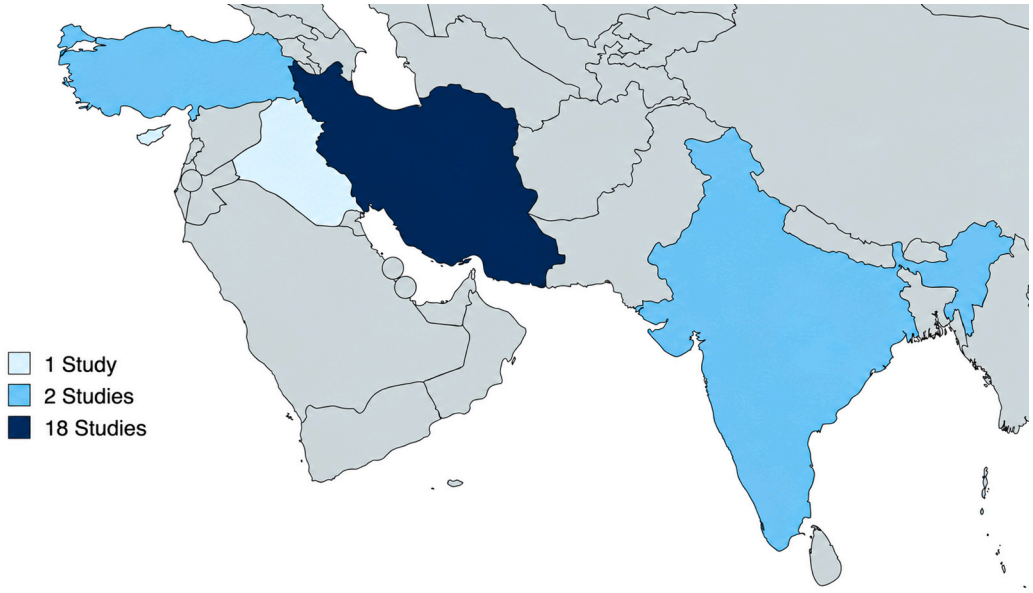


Fig. 2. Geographical distribution of included studies. Map showing the 24 unique studies by country: Iran (18), Turkey (2), India (2), Iraq (1), and Cyprus (1).

radiodermatitis trials tested multicomponent Alpha, henna-violet, or Dermolina-Henna formulations; Alpha improved healing, the henna-

Table 2
Study-level synthesis of comparative efficacy, safety reporting, and risk of bias.

Author (Year)	Regimen category	Principal comparative findings	Comparative interpretation	Safety assessment and findings	Risk of bias / quality appraisal
A — Sole-active henna (13 studies; 14 reports)					
Arslan et al.,[39]	A — Sole-active henna	Henna significantly improved most CIPN domains over time compared with control, with no significant effect on motor symptoms.	Favours henna for most CIPN domains; no clear motor-symptom difference.	AE assessment method NR; the authors reported observing no adverse effects from henna.	Some concerns
(Hekmatpou et al., 2018)	A — Sole-active henna	Henna significantly reduced the development of new pressure ulcers; Braden risk scores and temperature outcomes did not differ significantly.	Favours henna for incident pressure ulcers; other outcomes showed no clear difference.	NR.	Some concerns
Kayikci et al.,[43]	A — Sole-active henna	Henna did not significantly prevent capecitabine-induced hand–foot syndrome.	No clear between-group difference.	NR.	Serious risk
Keshavarz et al.,[44]	A — Sole-active henna	Henna ointment led to greater reduction in diaper dermatitis severity and erythema than hydrocortisone by day 5.	Favours henna preparation versus hydrocortisone.	Assessment method was not described; no significant side effects were reported in either group.	Low risk
Khosravi et al.,[45]	A — Sole-active henna	Wound depth declined within all three groups, and wound volume declined within the henna group. Although the authors described greater improvement with henna in the text, the tabulated tests were within-group and no pairwise between-group effect estimate or confidence interval was provided; comparative effect magnitude cannot be verified.	No between-group effect estimate was reported; no comparative conclusion.	NR.	Low risk
Lavari et al.,[56]	A — Sole-active henna	Henna oil significantly reduced pain intensity and disability compared with both sesame oil and placebo. SF-36 quality-of-life scores improved compared with placebo but not sesame oil, while diclofenac use did not differ significantly between groups.	Favours henna oil for pain and disability; quality-of-life findings were comparator-dependent.	Actively assessed using CTCAE v5.0. No serious adverse effects were observed. Mild local pruritus was reported by two henna participants and one placebo participant; no sesame-oil participant was reported with pruritus. Safety denominators were not specified.	Some concerns
Mohajerani et al.,[46]	A — Sole-active henna	No significant difference between henna hydrogel and placebo in treating hand-foot syndrome	No clear between-group difference.	Participants were asked to report adverse reactions. One henna-related allergic reaction led to exclusion. One noncompleter died; the cause and treatment relationship were not reported. No serious adverse effects were reported in the remaining participants.	High risk
Niazi et al.,[13]	A — Sole-active henna	Henna improved edema, itching, pain, sweating, and skin thinning versus placebo. Burning improvement was not significant, and skin redness increased, consistent with henna staining.	Favours henna for most symptoms; redness favoured placebo.	Participants were asked to report side effects, especially redness or allergy. No serious local or systemic adverse effects were reported; some placebo participants had worsening dermatitis, over-moistening, or skin fragility.	Low risk
Niazi et al.,[48]	A — Sole-active henna; single-arm	In the single-arm cohort, redness, itching, burning, and warmth improved from baseline; pain did not significantly change.	Improvement from baseline only; no comparative inference.	Assessment method was not described. One case of skin dryness was reported; no serious adverse events were reported.	Fair quality
Poursadra et al.,[28,29]	A — Sole-active henna; linked reports	Henna reduced ulcer area and PUSH scores versus standard care and olive oil, without significantly affecting body or ulcer temperature.	Favours henna for ulcer area and PUSH score; temperature outcomes showed no clear difference.	A 10-minute forearm allergy check was described before treatment; its result and later treatment-related adverse events were NR.	Some concerns (henna vs standard care); Moderate risk (henna vs olive oil)
Stavrinou et al.,[51]	A — Sole-active henna	Henna did not significantly reduce PPE-grade progression ($P = 0.063$), but improved HFS-related quality of life, pain, physical function, fatigue, global quality of life, and ECOG status.	Mixed: no clear PPE-progression difference; several symptom and quality-of-life outcomes favoured henna.	Assessment method was not described. No side effects were reported, and no participant discontinued because of the intervention.	Some concerns
Yaralizadeh et al.,[53]	A — Sole-active henna	Henna achieved complete fungal eradication, superior symptom resolution, greater Lactobacillus restoration, and better pH normalization than clotrimazole.	Favours henna preparation versus clotrimazole.	Actively monitored; no participant in either group reported a side effect.	Low risk
Zibanejad et al.,[54]	A — Sole-active henna	Henna and oak produced comparable improvements in wound healing and pain; both were superior to placebo or no intervention for the principal outcomes.	Favours henna versus placebo/no intervention; no clear difference versus oak.	Participants were instructed to stop treatment if irritation or dermatitis occurred; study-specific event counts were not reported.	Low risk

(*continued on next page*)

Table 2 (*continued*)

Author (Year)	Regimen category	Principal comparative findings	Comparative interpretation	Safety assessment and findings	Risk of bias / quality appraisal
B — Multicomponent henna-containing formulations (9 studies)					
Takkella et al.,[52]	B — Multicomponent mouthwash	The Lawsonia-containing mouthwash differed from distilled water for salivary AST, ALT, and LDH. The Punica mouthwash had lower LDH than the Lawsonia formulation; AST did not clearly differ. The source reports an ALT pairwise $P < 0.001$ despite almost identical final ALT means (22.6 vs 22.7).	Formulation-level evidence versus water; the henna-specific effect cannot be isolated. Interpret the reported ALT comparison with caution.	Immediate allergic reactions were observed for 1 h after rinsing; none occurred. Other adverse-event outcomes were not systematically reported.	Some concerns
Ansari et al.,[38]	B — Multicomponent formulation	Alpha® ointment accelerated overall radiodermatitis healing versus hydrocortisone and improved pain, pruritus, and discharge; burning relief did not differ significantly.	Favours Alpha formulation; effect is not henna-specific.	Assessment method was not described. No local or systemic reactions or worsening of dermatitis were reported among 60 participants.	High risk
Bhise et al.,[55]	B — Multicomponent formulation	Both the polyherbal NLC cream and clobetasol produced progressive reductions in PASI and itching scores. No between-group efficacy test was reported, so comparative effectiveness remains unclear.	No between-group efficacy test was reported; no comparative conclusion.	AEs were recorded at clinical follow-up. Two participants in the polyherbal-cream group reportedly withdrew because of itching; serious AEs were NR. Attrition reporting was internally inconsistent.	High risk
Dorbati et al.,[40]	B — Multicomponent formulation	Alpha® ointment improved wound-repair scores, selected REEDA components, and satisfaction. Pain findings were inconsistent: mean-rank pain favoured Alpha®, but the raw day-7 pain score was lower in the control group. Edema and ecchymosis did not differ.	Favours Alpha formulation for wound-repair measures; pain findings were inconsistent.	NR.	Some concerns
Elyasi et al.,[41]	B — Multicomponent formulation	Alpha® ointment did not prevent capecitabine-induced HFS but delayed severity progression over 4 chemotherapy courses.	No clear prevention effect; possible delay in severity progression.	Actively monitored; no treatment-related adverse reactions were reported.	High risk
Kamian et al.,[27]	B — Multicomponent formulation	Authors reported delayed grade 2–3 dermatitis and lower severity versus placebo; the published week-6 counts and percentages are inconsistent.	Favours the formulation, but not henna alone; inconsistent reporting limits confidence.	Monitored weekly; no allergic or other adverse effects reported. Safety denominator NR.	High risk
Khatami et al.,[57]	B — Multicomponent formulation	The CLA polyherbal cream produced significantly greater improvements in TIS and TEWL than the base cream. No significant between-group differences were found for itching/discomfort VAS, DLQI, stratum corneum hydration, or sebum production.	Favours CLA formulation for TIS and TEWL; other outcomes showed no clear difference.	Actively monitored using CTCAE v5.0. No serious adverse events were reported. One CLA participant discontinued after papular lesions; another discontinued because of pregnancy.	High risk
Pasalar et al.,[49]	B — Multicomponent formulation	Dermolina-Henna did not differ from mometasone for radiodermatitis grade, lesion number, erythema, or pain, but improved burning and itching.	Mixed: selected symptoms favoured Dermolina-Henna; other outcomes showed no clear difference.	Skin adverse events were monitored at follow-up. Adverse events led to withdrawal in 1 Dermolina-Henna participant and 3 mometasone participants (2.1% and 5.7%; between-group $P = 0.330$).	Some concerns
Salih et al.,[50]	B — Multicomponent formulation	The henna–tetracycline mixture had a first-session cure rate of 89% versus 94% with surgery ($P = 0.051$). with shorter treatment burden and fewer wound infections.	No clear difference in cure or recurrence; the combined regimen reduced treatment burden and wound infection. Effect is not henna-specific.	Reported complications: wound infection in 1/200 versus 8/200 participants and bleeding in 1/200 versus 3/200. A separate serious-adverse-event assessment was not reported.	High risk
C — Henna with a shared active cointervention (2 studies)					
Kakoei et al.,[42]	C — Shared active cointervention	No clear between-group difference in pain or clinical severity was reported between henna and chlorhexidine mouthwashes; both groups received corticosteroid therapy.	No clear difference between mouthwashes under shared corticosteroid therapy.	Adverse events were monitored, but study-specific results were not reported.	Low risk
Nabimeybodi et al.,[47]	C — Shared active cointervention	With antibiotic therapy shared by both groups, the henna-suppository add-on reduced vaginal discharge, cervical ulcer size, dyspareunia, and postcoital bleeding versus placebo; vulvovaginal irritation and friability did not differ.	Favours the henna add-on regimen for selected outcomes; not evidence for henna used alone.	Actively collected. One placebo participant discontinued because of antibiotic-related gastrointestinal complications; no henna-specific adverse event was reported.	Low risk

Note: Regimen categories are A, Lawsonia inermis as the sole stated active component; B, multicomponent henna-containing formulation; and C, henna evaluated alongside a shared active cointervention. Comparative findings and interpretations are based on between-group evidence. Within-group results are not used to infer comparative efficacy. "No between-group effect estimate" indicates that comparative efficacy could not be determined; single-arm evidence is non-comparative. NR means not reported and must not be interpreted as absence of adverse events. Poursadra and Rafiei are linked reports of an overlapping study and are counted once in the synthesis. Risk of bias was assessed using RoB 2 for randomized trials and ROBINS-I for nonrandomized comparative evidence; the NIH before–after quality assessment tool was used for the single-arm study.

Abbreviations: ALT, alanine aminotransferase; AST, aspartate aminotransferase; CIPN, chemotherapy-induced peripheral neuropathy; CLA, Curcuma longa, Lawsonia inermis, and Aristolochia rotunda; CTCAE, Common Terminology Criteria for Adverse Events; DLQI, Dermatology Life Quality Index; ECOG, Eastern Cooperative Oncology Group performance status; HFS, hand–foot syndrome; LDH, lactate dehydrogenase; NIH, National Institutes of Health; NLC, nanostructured lipid carrier; NR, not reported; PASI, Psoriasis Area and Severity Index; PPE, palmar-plantar erythrodysesthesia; PUSH, Pressure Ulcer Scale for Healing; REEDA, redness, edema, ecchymosis, discharge, and approximation; RoB 2, revised Cochrane risk-of-bias tool for randomized trials; ROBINS-I, Risk of Bias in Non-randomized Studies of Interventions; SF-36, 36-Item Short Form Health Survey; TEWL, transepidermal water loss; TIS, Three-Item Severity Score; VAS, visual analogue scale.

violet cream was reported to reduce dermatitis severity versus placebo, although week-six reporting was internally inconsistent, and Dermolina-Henna showed no clear difference in overall severity but improved selected symptoms.[27,38,49] Sole-active henna improved diaper-dermatitis severity versus hydrocortisone and improved most prosthesis-related contact-dermatitis symptoms versus vehicle placebo; however, redness favoured placebo and burning showed no clear difference.[13,44] The new CLA cream trial favoured its multicomponent formulation for TIS and transepidermal water loss, with no clear difference for itching, quality of life, hydration, or sebum.[57]

4.4.3. Pressure Ulcers

Pressure-ulcer severity reduction was described in two overlapping reports of one linked study family: henna reduced ulcer area and PUSH scores versus standard care and olive oil, while temperature measures did not clearly differ.[28,29] In a separate prevention trial, a new pressure ulcer developed in 1/37 henna participants and 6/35 controls by day 7; Braden risk scores and temperature measures did not clearly differ.[14]

4.4.4. Episiotomy wounds

For episiotomy wounds, Alpha improved selected repair and REEDA measures versus Betadine, but its pain results were inconsistent and it is a multicomponent product.[40] Sole-active henna improved healing and pain versus placebo or no treatment, with no clear difference from oak ointment.[54]

4.4.5. Pilonidal Sinus disease

In postoperative pilonidal disease, Khosravi et al.[45] reported within-group improvement but no between-group effect estimate, so comparative efficacy is not estimable.[45] Salih et al.[50] found no clear difference in cure or recurrence compared with surgery but less treatment burden and fewer wound infections for a henna–tetracycline mixture; this is a combined-regimen effect.[50]

4.4.6. Other conditions

Eight additional conditions were each assessed in a single study: chemotherapy-induced peripheral neuropathy, cervicitis, epidermolysis bullosa, chronic periodontitis, vaginal candidiasis, symptomatic oral lichen planus, chronic sciatica, and psoriasis.

For chemotherapy-induced peripheral neuropathy, Arslan et al. reported improvement in most sensory and functional outcomes versus routine care, but no clear motor-symptom difference.[39] In cervicitis, adding a henna vaginal preparation to antibiotics improved selected outcomes versus antibiotics plus placebo; this does not establish the effect of henna used alone.[47]

The single-arm epidermolysis bullosa study reported improvement from baseline in redness, itching, burning, and warmth, with no controlled comparison.[48] A multicomponent Lawsonia-containing periodontitis mouthwash produced mixed salivary biomarker findings; its effect cannot be attributed to henna alone, and the reported ALT pairwise comparison warrants caution.[52]

Henna cream produced more favourable reported vaginal-candidiasis outcomes than clotrimazole.[53] For oral lichen planus, henna and chlorhexidine mouthwashes showed no clear difference while both groups received corticosteroid treatment.[42] No clear difference should be described as equivalence.

For chronic sciatica, henna oil produced greater reductions in pain and disability than sesame oil and placebo; quality-of-life findings depended on the comparator.[56] In psoriasis, both the polyherbal cream and clobetasol groups improved, but no between-group efficacy test was reported, so comparative effectiveness is not estimable.[55]

4.5. GRADE certainty of evidence

Table 3 and Supplementary Table S9 present outcome-level GRADE assessments. Of 19 outcome groups, 12 were rated very low certainty and seven low certainty; none was rated moderate or high. The outcome-specific judgments reflect risk of bias, imprecision, and, where applicable, inconsistency or indirectness arising from multicomponent formulations and differing comparisons. These ratings limit confidence in apparent clinical benefits. For multicomponent and shared-treatment regimens, certainty applies to the evaluated regimen, not to henna alone.

4.6. Safety and adverse-event reporting

Across the 24 unique studies, 12 described ongoing adverse-event or complication surveillance, and four excluded participants with G6PD deficiency (Supplementary Table S10). Study-specific safety denominators were often unclear. Kamian et al. monitored participants weekly and reported no allergic or other cream-related adverse effects in either arm, but did not specify the safety denominator.[27] Reported events included an allergic reaction leading to exclusion in Mohajerani,[46] mild local pruritus in Lavari,[56] two itching-related withdrawals from the Bhise polyherbal-cream group,[55] a papular-lesion withdrawal from the Khatami CLA group,[57] and adverse-event withdrawals in one Dermolina-Henna and three mometasone participants in Pasalar.[49] In Mohajerani, one of the nine noncompleters died; the report did not state the cause of death or whether it was related to treatment.[46] In Salih, wound infection occurred in 1/200 versus 8/200 participants, and bleeding in 1/200 versus 3/200, for the henna–tetracycline regimen and surgery, respectively.[50] Serious events and reasons for withdrawal were inconsistently assessed or reported. Absence of reported events therefore does not establish general safety (Table 2; Supplementary Table S10).

5. Discussion

This review identified 24 unique studies of Lawsonia inermis-containing interventions across diverse clinical conditions. Some comparative outcomes favoured henna or a henna-containing formulation, but effects varied by condition, comparator, and preparation. Findings from multicomponent products and shared-treatment regimens cannot be assigned to henna alone. All 19 outcome groups had low or very low certainty, and incomplete adverse-event surveillance further limits clinical inference.

Existing reviews of henna have mainly addressed pharmacological or preclinical evidence. Batiha et al. described the various compounds found within Lawsonia inermis and highlighted its therapeutic potential.[2] Moutawalli et al. described antimicrobial, antitumour, antifungal, anti-inflammatory, and antioxidant properties of henna,[58] while Joyroy et al. reported antimicrobial and antioxidant activity of Lawsonia inermis extracts.[59] These pharmacological and preclinical findings provide biological plausibility for clinical investigation but do not establish the magnitude, certainty, or safety of treatment effects in humans.[15]

When compared with traditional plant-based therapies, the findings of this review can be considered within a broader ethnopharmacological context. Wound-related conditions such as episiotomy wounds, pressure ulcers, and pilonidal sinus disease have also been treated with plant-

Table 3
Direction of effect and certainty of evidence by clinical outcome and regimen category.

Clinical outcome	Studies, n (A/B/C)	Total N	Direction of effect	GRADE certainty	Reasons for downgrading
Radiodermatitis clinical severity reduction	3 B: Ansari (2013); Kamian (2022); Pasalar (2022)	196	Mixed: Ansari and Kamian reported benefit from multicomponent formulations, whereas Pasalar found no clear difference in radiodermatitis severity. Kamian's week-six data were internally inconsistent.	Very low ⊕◯◯◯	Very serious risk of bias, inconsistency, indirectness, and imprecision
Episiotomy wounds clinical severity reduction	2 A: Zibanejad (2020) B: Dorbati (2018)	288	Favours henna-containing regimens overall for wound-healing measures; findings were comparator-dependent and pain findings were mixed.	Low ⊕⊕◯◯	Serious indirectness and serious imprecision
Hand-foot syndrome prevention	2 A: Kayikci (2020) B: Elyasi (2022)	142	No clear comparative prevention effect; one multicomponent study suggested delayed severity progression.	Very low ⊕◯◯◯	Very serious risk of bias, serious indirectness, and very serious imprecision
Hand-foot syndrome clinical severity reduction	2 A: Mohajerani (2021); Stavrinou (2021)	65	No clear comparative difference in clinical severity.	Very low ⊕◯◯◯	Serious risk of bias and very serious imprecision
Pressure ulcer clinical severity reduction	1 A: Poursadra/Rafiei (2019)	108[c]	Favours sole-active henna for ulcer area and PUSH scores; temperature outcomes showed no clear difference.	Very low ⊕◯◯◯	Serious risk of bias and very serious imprecision
Pilonidal sinus disease treatment outcomes	1 B: Salih (2018)	400	No clear difference in cure or recurrence; treatment burden and wound infection favoured the combined henna-tetracycline regimen.	Very low ⊕◯◯◯	Very serious risk of bias, serious indirectness, and serious imprecision
Cervicitis clinical severity reduction	1 C: Nabimeybodi (2024)	82	Favours the henna add-on under shared antibiotics for selected outcomes; other outcomes showed no clear difference.	Low ⊕⊕◯◯	Very serious imprecision
Diaper dermatitis clinical severity reduction	1 A: Keshavarz (2016)	82	Favours sole-active henna preparation versus hydrocortisone.	Low ⊕⊕◯◯	Very serious imprecision
Chronic sciatica clinical severity reduction	1 A: Lavari (2023)	78	Favours sole-active henna oil for pain and disability; quality-of-life findings were comparator-dependent.	Very low ⊕◯◯◯	Serious risk of bias and very serious imprecision
Vaginal candidiasis clinical severity reduction	1 A: Yaralizadeh (2022)	80	Favours sole-active henna preparation versus clotrimazole.	Low ⊕⊕◯◯	Very serious imprecision
Prosthesis-related contact dermatitis clinical severity reduction	1 A: Niazi (2020)	74	Favours sole-active henna for most symptoms; burning showed no clear difference and redness favoured placebo.	Low ⊕⊕◯◯	Very serious imprecision
Pressure ulcer prevention	1 A: Hekmatpou (2018)	72	Favours sole-active henna added to routine care for incident pressure ulcers; other measures showed no clear difference.	Very low ⊕◯◯◯	Serious risk of bias and very serious imprecision
Contact dermatitis clinical severity reduction (multicomponent CLA formulation)	1 B: Khatami (2026)	62[b]	Favours the multicomponent CLA formulation for TIS and TEWL; other outcomes showed no clear difference.	Very low ⊕◯◯◯	Very serious risk of bias, serious indirectness, and very serious imprecision
Psoriasis clinical severity reduction (multicomponent NLC formulation)	1 B: Bhise (2022)	NR[a]	Comparative effect not estimable; no between-group efficacy test was reported.	Very low ⊕◯◯◯	Very serious risk of bias, serious indirectness, and very serious imprecision
Chemotherapy-induced peripheral neuropathy clinical severity reduction	1 A: Arslan (2020)	60	Favours sole-active henna for most CIPN domains; motor symptoms showed no clear difference.	Very low ⊕◯◯◯	Serious risk of bias and very serious imprecision
Chronic periodontitis salivary inflammatory biomarker reduction	1 B: Takkella (2023)	60	Mixed, with formulation-level evidence[e]: the Lawsonia-containing mouthwash differed from water for AST, ALT, and LDH; Punica had lower LDH. The reported ALT comparison is difficult to reconcile with nearly identical final means.	Very low ⊕◯◯◯	Serious risk of bias, serious indirectness, and very serious imprecision
Pilonidal sinus disease post-operative wound-healing	1 A: Khosravi (2025)	60	Comparative effect not estimable; no between-group effect estimate was reported.	Low ⊕⊕◯◯	Very serious imprecision
Symptomatic oral lichen planus clinical severity reduction	1 C: Kakoei (2022)	36	No clear difference between henna and chlorhexidine mouthwashes under shared corticosteroid therapy.	Low ⊕⊕◯◯	Very serious imprecision
Noncomparative evidence: epidermolysis bullosa clinical severity reduction	1 A: Niazi (2022); single-arm	7[d]	Improvement from baseline only; no comparative inference (single-arm).	Very low ⊕◯◯◯	Serious risk of bias and very serious imprecision

Notes: Study counts represent unique studies; linked reports were counted once. Regimen categories: A, Lawsonia inermis as the sole stated active component; B, multicomponent henna-containing formulation; and C, henna evaluated alongside a shared active cointervention. Direction-of-effect classifications were based on comparative between-group findings for clinically relevant outcomes and not on within-group change alone. GRADE ratings apply to the outcome-specific body of evidence for the evaluated henna-containing regimen(s). For categories B and C, conclusions apply to the combined formulation or add-on regimen and should not be attributed to henna alone. 'No clear difference' does not establish equivalence. The single-arm study was retained as noncomparative evidence. GRADE certainty was classified as high, moderate, low, or very low. [a]Bhise (2022): The article reports 65 enrolled participants, but group-allocation and attrition counts were inconsistent and the outcome-specific sample sizes were not reported. [b]Khatami (2026): The Results section and baseline table report 62 analyzed participants (30 CLA and 32 control), whereas the CONSORT diagram reports 60 participants (30 per group). [c]Poursadra and Rafiei were treated as linked reports of one overlapping study; N = 108 represents unique participants. [d]Niazi (2022) was a single-arm study; the reported improvement represents change from baseline rather than comparative efficacy. [e]Takkella (2023): The Lawsonia mouthwash also contained menthol and peppermint oil, which were absent from the water control. The source reports Punica versus Lawsonia ALT $P < 0.001$ although final mean ALT values were 22.6 versus 22.7; the contrast requires caution.

Abbreviations: ALT, alanine aminotransferase; AST, aspartate aminotransferase; CIPN, chemotherapy-induced peripheral neuropathy; CLA, Curcuma longa, Lawsonia inermis, and Aristolochia rotunda; CONSORT, Consolidated Standards of Reporting Trials; GRADE, Grading of Recommendations Assessment, Development and Evaluation; LDH, lactate dehydrogenase; N, number of participants included in the reported outcome analysis; NLC, nanostructured lipid carrier; NR, not reported; PASI, Psoriasis Area and Severity Index; PUSH, Pressure Ulcer Scale for Healing; TEWL, transepidermal water loss; TIS, Three-Item Severity Score.

based therapies including Aloe vera, turmeric, and neem, which have been investigated for their effects on tissue repair.[16–18] Likewise, Calendula officinalis and Matricaria chamomilla have been used for inflammatory skin conditions and have demonstrated anti-inflammatory and antioxidant properties.[19,20] These similarities provide contextual biological plausibility but do not establish that shared mechanisms account for the clinical findings observed with Lawsonia inermis. Differences in formulation, dose, comparator, and outcome assessment may also contribute to the observed findings.

In mucosal and infectious conditions, including vaginal candidiasis and cervicitis, findings should also be interpreted in the context of differences in formulation, concentration, comparator, and route of administration. Other plant-based therapies, including tea tree and garlic, contain compounds such as terpinen-4-ol and allicin with documented antimicrobial activity.[21,22] Antimicrobial activity has likewise been described for Lawsonia inermis[23]; however, these mechanistic similarities do not establish that they account for the clinical effects observed in the included studies.

Collectively, these observations demonstrate that *Lawsonia inermis* shares key phytochemical constituents, including flavonoids, phenolic compounds, and tannins, with a broader group of traditionally used medicinal plants targeting similar clinical conditions.[1,2,12] The overlap between traditional therapeutic indications and the clinical outcomes identified in this review strengthens the biological plausibility of *Lawsonia inermis*, while also highlighting areas where further clinical studies are required.

Recent systematic reviews of related complementary interventions further illustrate the need for cautious, outcome-specific interpretation of heterogeneous evidence. Lin et al.[60] evaluated plant-based topical agents for pressure-injury prevention, including a henna trial, and found that heterogeneity across plant preparations precluded meta-analysis, while methodological concerns limited confidence in the evidence.[60] Similarly, a systematic review of acupuncture for degenerative eye diseases reported favourable findings for selected outcomes but inconclusive effects for others.[61] These reviews provide a broader context for interpreting complementary therapies when interventions and outcomes are heterogeneous; they do not independently establish henna-specific efficacy or safety.

Henna may be accessible and inexpensive in some settings, but the current evidence does not establish that it can replace corticosteroids, surgery, antimicrobials, or established wound care. Formulation differences, uncertain comparative effects, and incomplete adverse-event surveillance require condition-specific evaluation before routine clinical use. Black-henna products containing added dyes raise separate concerns and should not be conflated with the preparations studied here.[62]

This review has several strengths, including its comprehensive synthesis of the available clinical literature on *Lawsonia inermis*. This review evaluates outcomes from a wide variety of medical specialties, providing a consolidated review that has not previously been done. Furthermore, the inclusion of randomized and non-randomized studies broadened the scope of evidence captured and helped identify areas requiring further research. This systematic review was prospectively registered in PROSPERO and conducted in accordance with the Cochrane Handbook (Chapter 12), SWiM reporting guidance, and PRISMA guidelines.[25,35,36] This review employed a predefined search strategy, duplicate screening, the use of verified risk of bias tools (RoB 2, ROBINS-I, and NIH), and the use of the GRADE framework for assessing the certainty of evidence. These prespecified methods and transparent study-level assessments allow readers to inspect the methodological judgments and the limits of inference. Lastly, this review focused on important clinical outcomes rather than preclinical or phytochemical evidence, intending to bridge the gap between herbal medicine and modern medical practices. The findings of this study fall within the established pharmacological literature available on henna, strengthening clinical plausibility but not certainty.

Several limitations should be considered when interpreting this review. Considerable clinical and methodological heterogeneity was present across studies, including variability in formulation, concentration, treatment regimens, comparators, outcome measures, and follow-up durations. This heterogeneity limited comparability across studies and prevented meta-analysis. The certainty of evidence was low or very low across all outcome groups, with frequent downgrading for risk of bias and imprecision. Several studies had methodological concerns related to blinding, outcome assessment, and deviations from intended interventions. The Poursadra/Rafiei reports described overlapping cohorts, and some studies reported inconsistent or incomplete participant and analysis denominators.[28,29] Several outcomes also lacked between-group estimates or confidence intervals, limiting comparative interpretation. Eligibility was restricted to English-language full-text publications, and unpublished evidence was not comprehensively searched. One eligible report was found through an author search outside the database searches, so other eligible studies may also have been missed. In the Kamian trial, exclusions after randomization and internally inconsistent week-six dermatitis counts and percentages further limit interpretation of its reported benefit.[27] Geographic concentration in a limited number of countries may further restrict generalizability. Finally, few studies provided a clear safety denominator or systematic long-term adverse-event follow-up, limiting conclusions regarding safety and tolerability.

Future research should evaluate clearly defined henna preparations and doses using clinically appropriate comparators and prespecified patient-important outcomes. Trials should report between-group effect estimates with confidence intervals rather than relying on within-group changes alone. Studies should also distinguish efficacy and safety denominators and incorporate prospective adverse-event surveillance, including serious events, allergic or skin reactions, treatment withdrawals, and G6PD-related precautions where relevant.

6. Conclusion

Lawsonia inermis-containing interventions showed favourable findings for selected outcomes, but effects from combined formulations and add-on regimens cannot be attributed to henna alone. Across 19 outcome groups, certainty was low or very low, and some studies did not report a comparative estimate. Adverse-event surveillance was insufficient to establish general safety. The evidence does not support replacing established care; better reported comparative trials and systematic safety assessment are needed.

Abbreviations

AE, adverse event; **ALT**, alanine aminotransferase; **CENTRAL**, Cochrane Central Register of Controlled Trials; **CIPN**, chemotherapy-induced peripheral neuropathy; **CIPNAT**, Chemotherapy-Induced Peripheral Neuropathy Assessment Tool; **CLA**, Curcuma longa, Lawsonia inermis, and Aristolochia rotunda; **G6PD**, glucose-6-phosphate dehydrogenase; **GRADE**, Grading of Recommendations Assessment, Development and Evaluation; **HFS**, hand–foot syndrome; **ICU**, intensive care unit; **IL-1β**, interleukin-1 beta; **MPNS**, Medicinal Plant Names Services; **NF-κB**, nuclear factor kappa B; **NIH**, National Institutes of Health; **NSAIDs**, nonsteroidal anti-inflammatory drugs; **PICO**, Population, Intervention, Comparator, and Outcomes; **PRISMA**, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; **PROSPERO**, International Prospective Register of Systematic Reviews; **PUSH**, Pressure Ulcer Scale for Healing; **QoL**, quality of life; **RCT**, randomized controlled trial; **REEDA**, redness, edema, ecchymosis, discharge, and approximation; **ROBINS-I**, Risk of Bias in Non-randomized Studies of Interventions; **RoB 2**, revised Cochrane risk-of-bias tool for randomized trials; **RTOG**, Radiation Therapy Oncology Group; **SWiM**, Synthesis Without Meta-analysis; **TIS**, Three-Item Severity Score; **TNF-α**, tumor necrosis factor alpha; **VAS**, visual analogue scale.

CRediT authorship contribution statement

Zaid Chilmeran: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ahmed Mahdi:** Validation, Investigation, Formal analysis, Data curation. **Abdullah Hussain:** Writing – review & editing, Writing – original draft, Investigation. **Abu-baker Sharafeldin:** Writing – original draft, Investigation. **Salim Fredericks:** Writing – review & editing, Supervision.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ctim.2026.103470.

References

1. Chaudhary G, Goyal S, Poonia P. Lawsonia inermis Linnaeus: a phytopharmacological review. *Int J Pharm Sci Drug Res*. 2010;2:91–98.
2. Batiha GE-S, Teibo JO, Shaheen HM, et al. Therapeutic potential of Lawsonia inermis Linn: a comprehensive overview. *Naunyn-Schmiede's Arch Pharmacol*. 2024;397:3525–3540.
3. Díaz-Herrera MÁ, Martínez-Riera JR, Verdú-Soriano J, et al. Multicentre study of chronic wounds point prevalence in primary health care in the southern metropolitan area of Barcelona. *J Clin Med*. 2021;10:797.
4. Guest JF, Ayoub N, Mcilwraith T, et al. Health economic burden that wounds impose on the National Health Service in the UK. *BMJ Open*. 2015;5, e009283.
5. Heyer K, Herberger K, Protz K, Glaeske G, Augustin M. Epidemiology of chronic wounds in Germany: analysis of statutory health insurance data. *Wound Repair Regen*. 2016;24:434–442.
6. Martinengo L, Olsson M, Bajpai R, et al. Prevalence of chronic wounds in the general population: systematic review and meta-analysis of observational studies. *Ann Epidemiol*. 2019;29:8–15.
7. Nussbaum SR, Carter MJ, Fife CE, et al. An economic evaluation of the impact, cost, and medicare policy implications of chronic nonhealing wounds. *Value Health*. 2018;21:27–32.
8. Yasir, M., Goyal, A. & Sonthalia, S. 2018. Corticosteroid adverse effects.
9. Niculet E, Bobeica C, Tatu AL. Glucocorticoid-induced skin atrophy: the old and the new. *Clin Cosmet Investig Dermatol*. 2020:1041–1050.
10. Ozawa S, Shankar R, Leopold C, Orubu S. *Access to medicines through health systems in low-and middle-income countries*. Oxford University Press; 2019.
11. Hidayat NN, Musfiroh I. Lawsonia inermis: Review of Its Anti-inflammatory Properties and Therapeutic Potential. *Indones J Pharm Sci Technol*. 2025;12:81–89.
12. Sattar S, Shabbir A, Shahzad M, et al. Evaluation of anti-inflammatory and immunomodulatory potential of Lawsone (2-hydroxy-1, 4-naphthoquinone) using pre-clinical rodent model of rheumatoid arthritis. *Front Pharmacol*. 2023;14, 1279215.
13. Niazi M, Mehrabani M, Namazi MR, et al. Efficacy of a topical formulation of henna (Lawsonia inermis L.) in contact dermatitis in patients using prosthesis: A double-blind randomized placebo-controlled clinical trial. *Complement Ther Med*. 2020;49, 102316.
14. Hekmatpou D, Ahmadian F, Eghbali M, Farsaei S. Henna (Lawsonia inermis) as an inexpensive method to prevent decubitus ulcers in critical care units: A randomized clinical trial. *J Evid-Based Integr Med*. 2018;23, 2515690X18772807.
15. Joshi C, Shukla K, Odedra KN, Jadeja B. Lawsonia inermis L.(Henna): A Comprehensive Review of Its Phytochemistry, Pharmacological Potential, Traditional Uses, and Commercial Applications. *J Plant Biota*. 2025.
16. Surjushe A, Vasani R, Saple D. Aloe vera: a short review. *Indian J Dermatol*. 2008;53:163–166.
17. Hewlings SJ, Kalman DS. Curcumin: A review of its effects on human health. *Foods*. 2017;6:92.
18. Subapriya R, Nagini S. Medicinal properties of neem leaves: a review. *Curr Med Chem-Anti-Cancer Agents*. 2005;5:149–156.
19. Mckay DL, Blumberg JB. A review of the bioactivity and potential health benefits of chamomile tea (Matricaria recutita L.). *Phytother Res Int J Devoted Pharmacol Toxicol Eval Nat Product Deriv*. 2006;20:519–530.
20. Preethi KC, Kuttan R. Wound healing activity of flower extract of Calendula officinalis. *J Basic Clin Physiol Pharmacol*. 2009;20:73–79.
21. Borlinghaus J, Albrecht F, Gruhlke MC, Nwachukwu ID, Slusarenko AJ. Allicin: chemistry and biological properties. *Molecules*. 2014;19:12591–12618.
22. Carson CF, Hammer KA, Riley TV. Melaleuca alternifolia (tea tree) oil: a review of antimicrobial and other medicinal properties. *Clin Microbiol Rev*. 2006;19:50–62.
23. Muhammad H, Muhammad S. The use of Lawsonia inermis Linn.(henna) in the management of burn wound infections. *Afr J Biotechnol*. 2005;4.
24. Grzanna R, Lindmark L, Frondoza CG. Ginger—an herbal medicinal product with broad anti-inflammatory actions. *J Med Food*. 2005;8:125–132.
25. Page MJ, Mckenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *bmj*. 2021:372.
26. Lesan M, Kamian S, Heidarirad G, Mehriardestani M. 16P Topical henna cream in prevention of radiation-induced dermatitis in breast cancer: A randomized double-blind clinical trial. *Ann Oncol*. 2020;31. S1248.
27. Kamian S, Heydarirad G, Mehraardestani M, Lesan M. Effects of Henna-Violet Based Topical Preparation in Preventing and Reducing the Severity of Radiation-Induced Dermatitis in Breast Cancer Patients: A Randomized, Double-Blind Clinical Trial. *Nov Biomed*. 2022;10:246–254.
28. Poursadra E, Anvari-Tafti M, Dehghani A, Eghbali-Babadi M, Rafiei Z. Comparing the effect of henna oil and olive oil on pressure ulcer grade one in intensive care units patients. *Adv Biomed Res*. 2019;8:68.
29. Rafiei Z, Mazaheri M, Eghbali-Babadi M, Yazdannik A. The effect of henna (Lawsonia inermis) on preventing the development of pressure ulcer grade one in intensive care unit patients. *Int J Prev Med*. 2019;10:26.
30. Sterne JA, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. *bmj*. 2016;355.
31. Sterne JA, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *bmj*. 2019;366.
32. National Heart, L. & Institute, B. 2019. Study Quality Assessment Tools-Quality Assessment Tool for Before-After (Pre-Post) Studies With No Control Group.
33. Guyatt GH, Oxman AD, Vist GE, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. *Bmj*. 2008;336:924–926.
34. Royal Botanic Gardens K. *Medical Plant Names Services (MPNS) [Online]*. Kew, UK: Royal Botanic Gardens, Kew; 2026. https://mpns.science.kew.org/. Accessed April 2, 2026].
35. Campbell M, Mckenzie JE, Sowden A, et al. Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline. *bmj*. 2020;368.
36. Jpt H. *Cochrane Handb Syst Rev Interv*; 2008. http://www.cochrane-handbook.org.
37. Yucel I, Guzin G. Topical henna for capecitabine induced hand–foot syndrome. *Investig New Drugs*. 2008;26:189–192.
38. Ansari M, Dehsara F, Mosalaei A, Omidvari S, Ahmadloo N, Mohammadianpanah M. Efficacy of topical alpha ointment (containing natural henna) compared to topical hydrocortisone (1%) in the healing of radiation-induced dermatitis in patients with breast cancer: a randomized controlled clinical trial. *Iran J Med Sci*. 2013;38:293.
39. Arslan S, Bahceli PZ, Ilik Y, Artaç M. The preliminary effects of henna on chemotherapy-induced peripheral neuropathy in women receiving oxaliplatin-based treatment: A parallel-group, randomized, controlled pilot trial. *Eur J Oncol Nurs*. 2020;48, 101827.
40. Dorbati PN, Mahmoodi Z, Salehi K, Dolatian M, Mahmoodi A. The effects of alpha® ointment (containing natural Henna) and Betadine® solution on Episiotomy healing in Primiparous women: a randomized controlled trial. *Iran Red Crescent Med J*. 2018;20.
41. Elyasi S, Rasta S, Taghizadeh-Kermani A, Hosseini S. Topical henna and curcumin (Alpha®) ointment efficacy for prevention of capecitabine induced hand-foot syndrome: a randomized, triple-blinded, placebo-controlled clinical. *DARU J Pharm Sci*. 2022;30:117–125.
42. Kakoei S, Karbasi N, Raeiszadeh M, Tajadini H, Nekouei AH. The efficacy of henna (Lawsonia inermis L.) mouthwash versus chlorhexidine gluconate 0.2% mouthwash as adjuvant therapy of oral lichen planus: A randomized double-blind clinical trial. *J Ethnopharmacol*. 2022;290, 115037.
43. Kayikci EE, Can G, Sen F, Saip P. Henna Application in the prevention of capecitabine-induced hand-foot syndrome in breast and colorectal cancer patients. *Florence Nightingale J Nurs*. 2020;28:299.
44. Keshavarz A, Zeinaloo AA, Mahram M, Mohammadi N, Sadeghpour O, Maleki MR. Efficacy of traditional medicine product henna and hydrocortisone on diaper dermatitis in infants. *Iran Red Crescent Med J*. 2016;18, e24809.
45. Khosravi SA, Kamalinejad M, Javadi SM, Namdar H, Fesharaki MG. Wound-Healing Properties of Henna in Pilonidal Sinus Surgery: A Randomized Study. *Indian J Dermatol*. 2025;70:221.
46. Mohajerani R, Shahi F, Jafariazar Z, Afshar M. Efficacy of topical Lawsonia inermis L.(Henna) hydrogel in fluorouracil-induced hand-foot syndrome: a pilot randomized double-blind placebo-controlled clinical trial. *Cutan Ocul Toxicol*. 2021;40:257–262.
47. Nabimeybodi N, Nokhostin F, Zareshahi R, et al. The effect of Henna (Lawsonia inermis) vaginal suppository combined with antibiotic therapy in the treatment of cervicitis: An RCT. *Int J Reprod Biomed*. 2024;22:269.
48. Niazi M, Parvizi MM, Saki N, Parvizi Z, Mehrbani M, Heydari M. Efficacy of a topical formulation of henna (Lawsonia inermis Linnaeus) on the itch and wound healing in patients with epidermolysis bullosa: A pilot single-arm clinical trial. *Dermatol Pract & Concept*. 2022;12, e2022115.
49. Pasalar M, Ahadi B, Mirzaei HR, et al. Comparing dermolina-henna cream with mometasone cream in improving radiodermatitis among patients with breast cancer:

a randomized active-control double-blind clinical trial. *J Integr Complement Med*. 2022;28:895–903.
50. Salih AM, Kakamad FH, Salih RQ, et al. Nonoperative management of pilonidal sinus disease: one more step toward the ideal management therapy—a randomized controlled trial. *Surgery*. 2018;164:66–70.
51. Stavrinou M, Tsitsi T, Astras G, Paikousis L, Charalambous A. A randomised controlled feasibility trial to evaluate Lawsonia inermis (henna)'s effect on palmar-plantar erythrodysesthesia induced by capecitabine or pegylated liposomal doxorubicin. *Eur J Oncol Nurs*. 2021;51, 101908.
52. Takkella BK, Anusha NV, Balaji DL, et al. The Comparison of the Anti-inflammatory Efficacy of Phytochemical Extracts in Punica granatum and Lawsonia inermis Among Patients Diagnosed With Chronic Periodontitis. *Cureus*. 2023;15.
53. Yaralizadeh, M., Abedi, P., Nezamivand, C.S., Yaralizadeh, M., Hormozie, L., Hejazi, Z.A. & Ghanbari, S. 2022. Comparing the effects of two cream formulations, Henna (Lawsonia inermis) and clotrimazole, on vaginal candidiasis: A randomized controlled trial.
54. Zibanejad S, Miraj S, Kopaei MR. Healing effect of Quercus persica and Lawsonia inermis ointment on episiotomy wounds in primiparous women. *J Res Med Sci*. 2020; 25:11.
55. Bhise K, Mulla G, Shaikh S, Ziya-Ul-Haq-Chowdhary KS, Khan S. An open-label randomized clinical study of topical nanoparticulate antipsoriatic polyherbal cream. *Asian J Pharm Clin Res*. 2022;15:139–144.
56. Lavari N, Ostadrahimi N, Rahimi R, Raei M, Abbassian A. The effect of a topical formulation from Lawsonia inermis L.(henna) on pain intensity in patients with chronic sciatica: A randomized double-blind clinical trial. *J Ethnopharmacol*. 2023; 313, 116519.
57. Khatami S, Sadati Lamardi SN, Najafi-Vosough R, Mizani A, Shirbeigi L, Abbassian A. The Effectiveness of a Topical Turmeric-Based Formulation on Contact Dermatitis: A Randomized, Double-Blind, Controlled Trial. *Iran J Pharm Res*. 2026; 25, e172121.
58. Moutawalli A, Benkhouili FZ, Doukkali A, Benzeid H, Zahidi A. The biological and pharmacologic actions of Lawsonia inermis L. *Phytomedicine*. 2023;3, 100468.
59. Joyroy N, Ngiwsara L, Wannachat S, Mingma R, Svasti J, Wongchawalit J. Unveiling the potentials of Lawsonia inermis L.: its antioxidant, antimicrobial, and anticancer potentials. *PeerJ*. 2025;13, e19170.
60. Lin HC, Yang LY, Su YC, Lee BO. The effectiveness of plant-based topical agents for the prevention of pressure injuries: Systematic review of randomised controlled trials. *Int J Nurs Stud*. 2025;167, 105069.
61. Chen KY, Chan HC, Chan CM. Is acupuncture a viable therapeutic strategy for degenerative eye diseases? a systematic review and meta-analysis. *Complement Ther Med*. 2025;93, 103235.
62. FDA. *Tempor Tattoos Henna/Mehndi "Black Henna" Fact Sheet [Online]*; 2024. https://www.fda.gov/cosmetics/cosmetic-products/temporary-tattoos-hennamehndi-and-black-henna-fact-sheet (Available).